BTECH COMPUTER SCIENCE • DATA SCIENCE

FRAUDLENS

DIGITAL PAYMENT FRAUD DETECTION USING EXPLAINABLE MACHINE LEARNING

VISUAL PROTOTYPE & INTERFACE WALKTHROUGH

A polished interface concept for transaction monitoring, risk review, and explainability.



Prototype status: Screens use synthetic sample records, illustrative heuristic risk scores, and conceptual feature-impact graphics. They are not real model predictions or SHAP outputs.

01 | Prototype Overview

FraudLens is a web-interface prototype for the digital payment fraud-detection project. It presents a guided workflow for viewing transaction activity, inspecting an individual transaction, exploring records, and understanding how model explanations will be displayed after integration.

DASHBOARD	TRANSACTION ANALYSIS	EXPLORER
Summary indicators, activity chart, risk distribution, and recent records.	Transaction inputs, illustrative risk category, and transparent rule reasons.	Filters, transaction table, risk bands, and CSV export flow.
EXPLAINABILITY	**HUMAN REVIEW**	**CURRENT LIMIT**
Conceptual feature-impact area; genuine SHAP outputs are pending.	High-risk cases are framed for investigation, not automatic accusation.	Trained model and real SHAP values are not connected in this UI yet.

Design goals

- Consistent dark theme and clear navigation
- High-contrast risk states and concise labels
- Transaction-level details without clutter
- Clear disclosure of demo-only data and scores
- Integration path for model inference and SHAP explanations

02 | Overview Dashboard

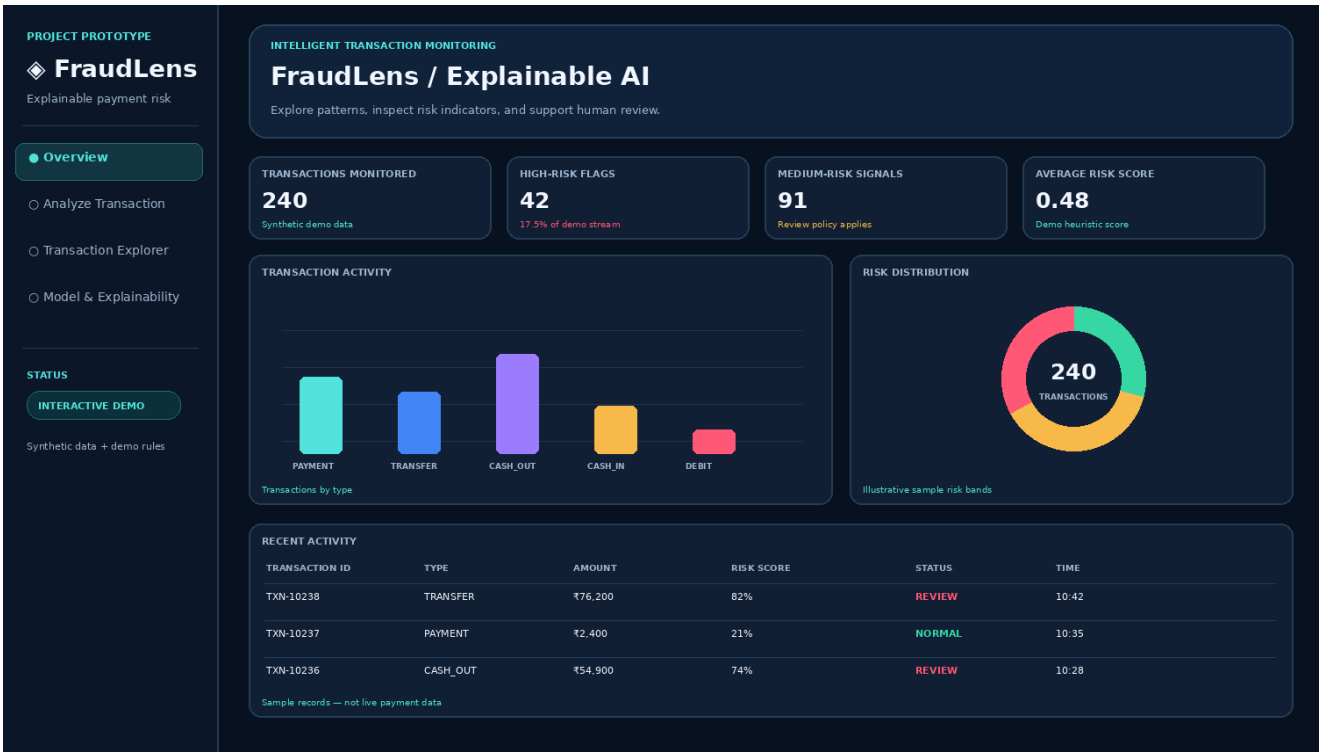


Figure 1. Dashboard mockup. Summary values and transaction rows are sample data used to demonstrate layout.

03 | Analyze Transaction

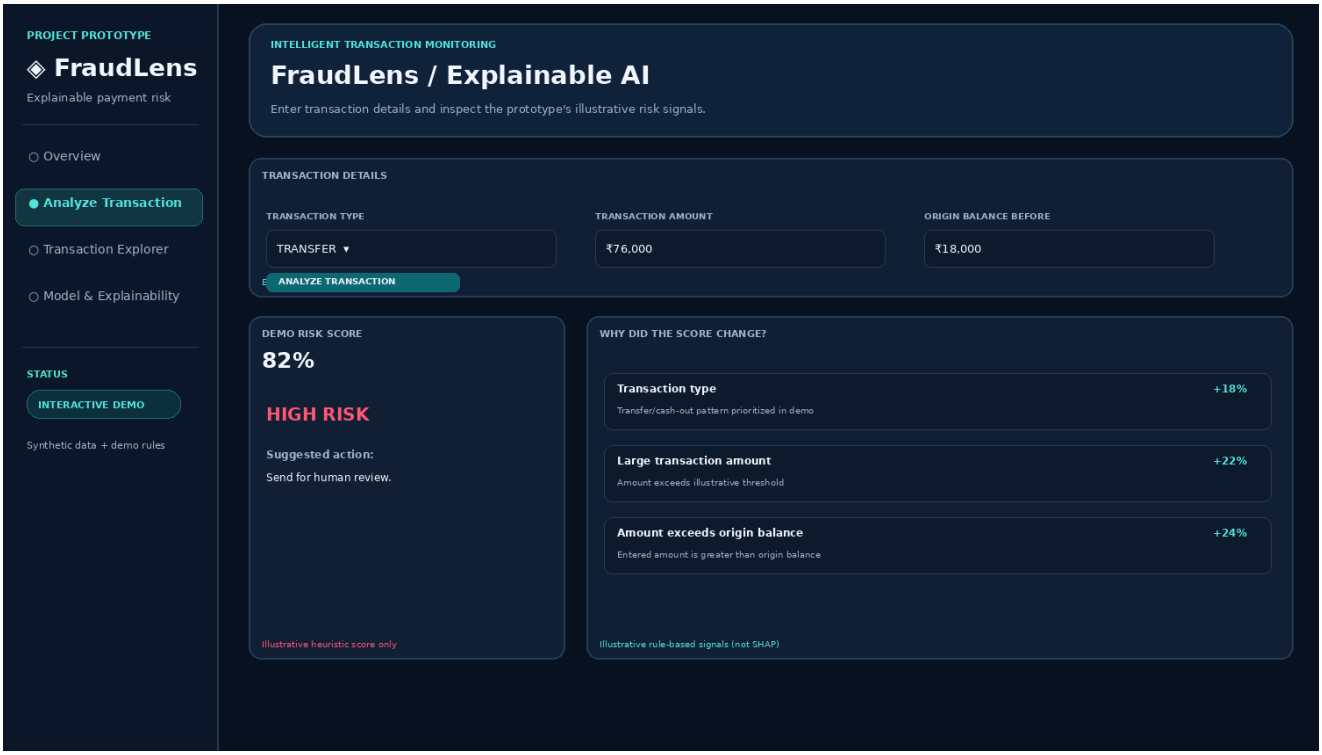


Figure 2. Analysis mockup. Risk score and rule contributions are illustrative; they are not trained-model predictions.

User flow

1. Select transaction type and enter amount/balance fields.
2. Run the demo analysis.
3. Review the risk band and rule-based signals.
4. Treat the suggested action as a prompt for human review only.

04 | Transaction Explorer

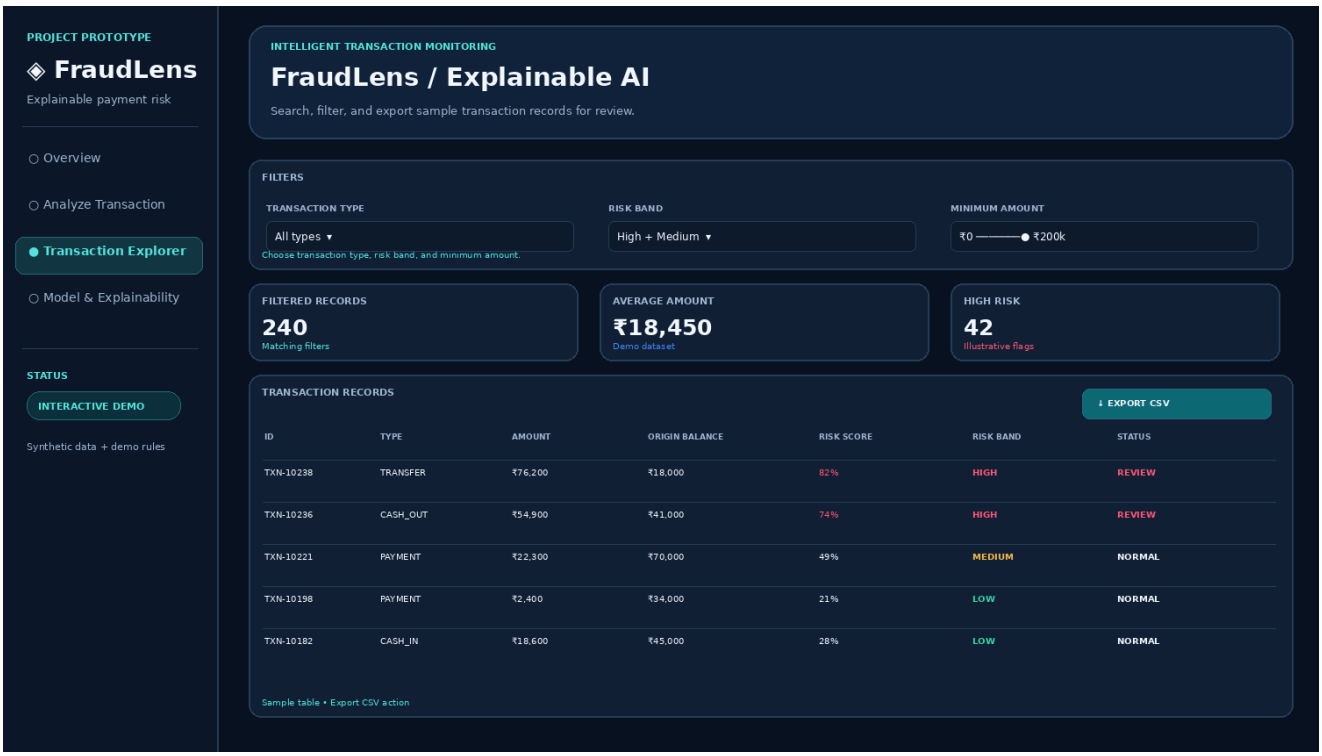


Figure 3. Explorer mockup with filters, summary cards, sample records, and CSV export action.

05 | Model & Explainability

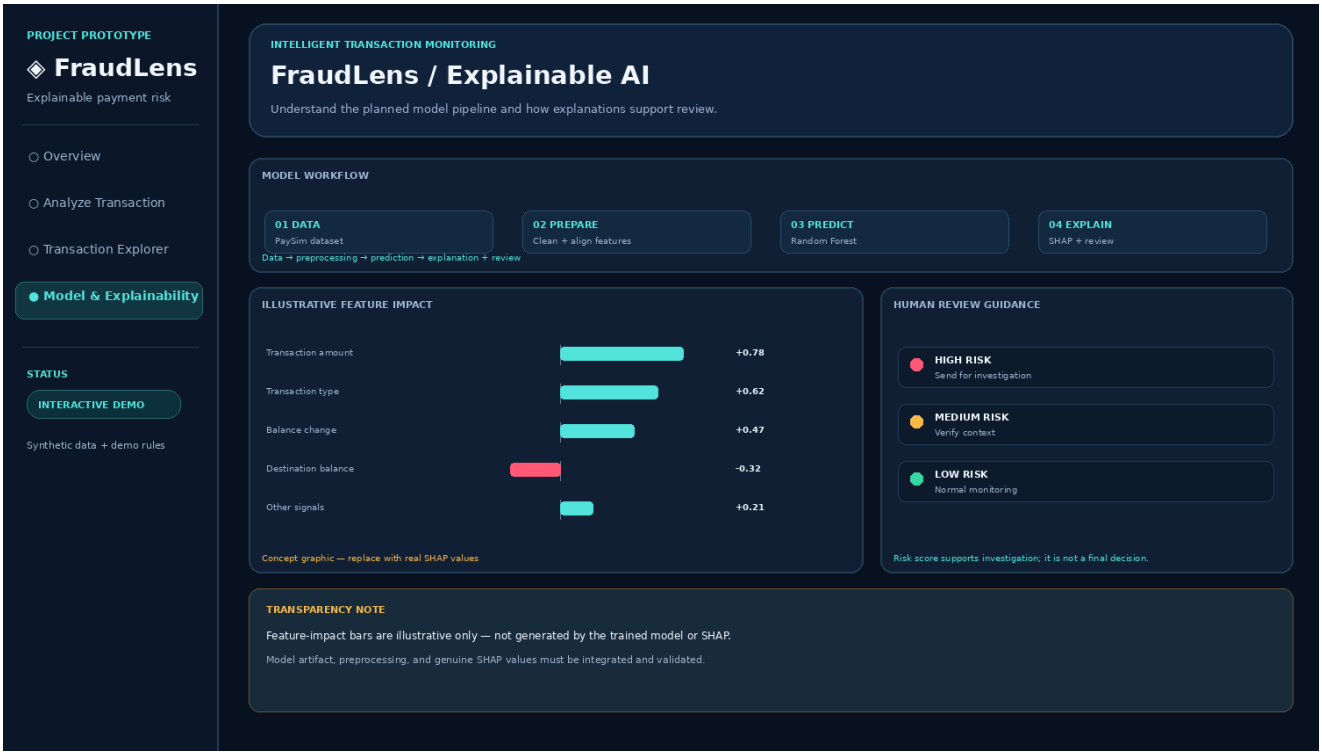


Figure 4. Explainability mockup. Feature-impact bars are conceptual and must be replaced with real SHAP values from the trained model.

06 | Technical Workflow

STAGE	PURPOSE	CURRENT STATUS
Data	Use PaySim or approved transaction data.	Synthetic sample records in UI.
Preprocessing	Match training features, encoding, and column order.	Saved pipeline not connected.
Prediction	Run trained classifier and selected threshold.	Heuristic demo scoring used.
Explainability	Generate prediction-linked SHAP values.	Conceptual visual; integration pending.
Human review	Review flagged transactions and document decisions.	Review guidance shown conceptually.

07 | Demonstration Checklist

- Open Overview and explain that KPI cards use synthetic values.
- Analyze a sample transaction and explain the rule-based result.
- Demonstrate Explorer filters and CSV export.
- Explain that the feature-impact graphic is conceptual, not SHAP output.
- Describe the next milestone: connect the trained model, align preprocessing, and validate on held-out data.

Next implementation steps

1. Load the trained model artifact and exact preprocessing pipeline.
2. Align entered features with training columns.
3. Replace heuristic scoring with model inference.
4. Generate genuine SHAP explanations for model predictions.
5. Test errors and document reproducible metrics.

Academic integrity note
These are designed UI mockups, not screenshots captured from a running app. Sample figures and risk outputs are illustrative and must not be presented as model evidence.